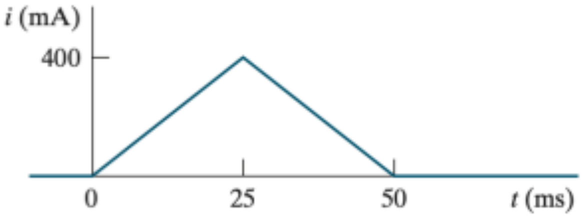
6.7 The triangular current pulse shown in Fig. P6.7 is applied to a 375 mH inductor.

PSPICE MULTISIM

a) Write the expressions that describe $i(t)$ in the four intervals $t < 0$, $0 \leq t \leq 25$ ms, $25 \text{ ms} \leq t \leq 50$ ms, and $t > 50$ ms.

b) Derive the expressions for the inductor voltage, power, and energy. Use the passive sign convention.

Figure P6.7



6.37 a) Show that the differential equations derived in (a) of Example 6.8 can be rearranged as follows:

$$4\frac{di_1}{dt} + 25i - 8\frac{di_2}{dt} - 20i_2 = 5i_g - 8\frac{di_g}{dt};$$

$$-8\frac{di_1}{dt} - 20i_1 + 16\frac{di_2}{dt} + 80i_2 = 16\frac{di_g}{dt}.$$

$$4\frac{di_1}{dt} + 8\frac{d}{dt}(i_g - i_2) + 20(i_1 - i_2) + 5(i_1 - i_g) = 0.$$

$$20(i_2 - i_1) + 60i_2 + 16\frac{d}{dt}(i_2 - i_g) - 8\frac{di_1}{dt} = 0.$$

$$i_g = 16 - 16e^{-5t}\ \text{A},$$

b) Show that the solutions for i_1 and i_2 given in (b) of Example 6.8 satisfy the differential equations given in part (a) of this problem.

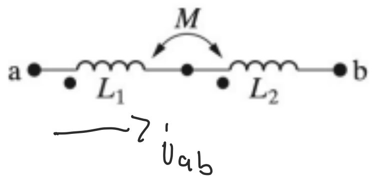
$$i_1 = 4 + 64e^{-5t} - 68e^{-4t}\ \text{A},$$

$$i_2 = 1 - 52e^{-5t} + 51e^{-4t}\ \text{A}.$$

6.38 a) Show that the two coupled coils in Fig. P6.38 can be replaced by a single coil having an inductance of $L_{ab} = L_1 + L_2 + 2M$. (*Hint:* Express v_{ab} as a function of i_{ab}.)

b) Show that if the connections to the terminals of the coil labeled L_2 are reversed, $L_{ab} = L_1 + L_2 - 2M$.

Figure P6.38

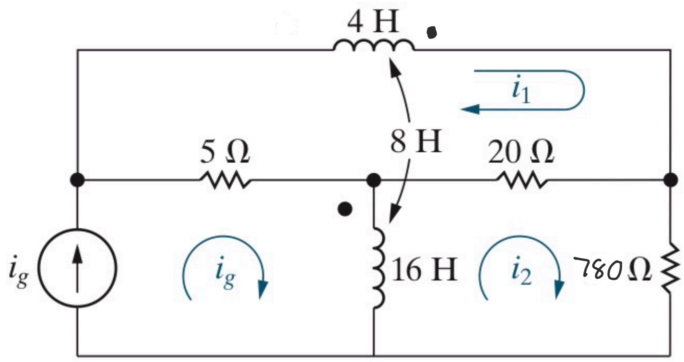


6.41 a) Write a set of mesh-current equations for the circuit in Example 6.8 if the dot on the 4 H inductor is at the right-hand terminal, the reference direction of i_g is reversed, and the 60 Ω resistor is increased to 780 Ω.

b) Verify that if there is no energy stored in the circuit at $t = 0$, and if $i_g = 1.96 - 1.96e^{-4t}$ A, the solutions to the differential equations derived in part (a) are

$$i_1 = -0.4 - 11.6e^{-4t} + 12e^{-5t} \text{ A},$$

$$i_2 = -0.01 - 0.99e^{-4t} + e^{-5t} \text{ A}.$$



6.43 There is no energy stored in the circuit in Fig. P6.43 at the time the switch is opened.

a) Derive the differential equation that governs the behavior of i_2 if $L_1 = 10$ H, $L_2 = 40$ H, $M = 5$ H, and $R_o = 90\ \Omega$.

b) Show that when $i_g = 10e^{-t} - 10$ A, $t \geq 0$, the differential equation derived in (a) is satisfied when $i_2 = e^{-t} - 5e^{-2.25t}$ A, $t \geq 0$.

c) Find the expression for the voltage v_1 across the current source.

d) What is the initial value of v_1? Does this make sense in terms of known circuit behavior?

Figure P6.43